Assignment 2

Unit: KIT717 Internet of Things and Web Applications

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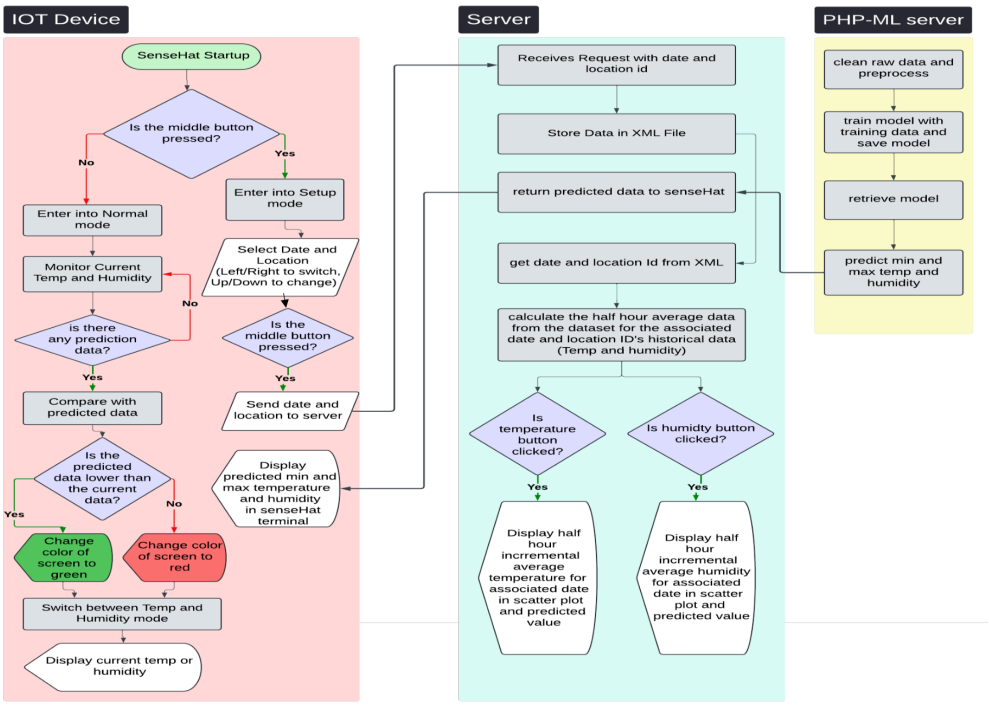
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Flow Chart Diagram:

Diagram explaining data flow in the system

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Diagram explaining data flow in the system



Explanation of Machine Learning Algorithm Choice

For this project, supervised linear regression was selected as the machine learning algorithm. The decision was based on the nature of the prediction task: weather conditions follow seasonal patterns that can be reasonably captured by a linear model when the input features are chosen appropriately. Linear regression is simple to implement, easy to interpret, and well-suited for continuous output variables such as temperature and humidity.

The input features were engineered to represent seasonality and location. Rather than using a raw timestamp — which would imply a linear trend over time rather than a repeating cycle — the day-of-year is encoded as sine and cosine values. This maps any day onto a unit circle so that the end of one year connects smoothly to the start of the next. Site identity is represented using one-hot encoding so that each location is treated as a separate category without implying any numerical ordering between sites.

The feature vector used for each prediction is:

```
[ sin(2π · day_of_year / 365),  cos(2π · day_of_year / 365),
is_site1,  is_site2,  is_site3,  is_site4,  is_site5 ]
```

Four separate models are trained — one each for maximum temperature, minimum temperature, maximum humidity, and minimum humidity — using PHP-ML's LeastSquares implementation. Humidity predictions are clamped to the range [0, 100] after inference, since linear regression can produce values outside physically valid bounds.

Training Data and Sampling Steps

The training data was sourced from a historical weather dataset recorded at five Tasmanian monitoring sites, containing half-hourly readings of temperature and humidity. The dataset originally spanned several years across five locations.

The initial dataset contained inconsistencies, missing values, and unnecessary columns that were removed during the data cleaning process. The final cleaned dataset consisted of four fields: location ID, date-time, humidity, and temperature. Missing data points were averaged with adjacent observations and integrated back into the dataset. Temperature values of 0 were retained as they can legitimately occur. Instances where humidity was recorded as 100% were corrected by averaging the preceding and subsequent values.

To train the models, the CSV data was aggregated to one row per (date, site) pair by computing the daily minimum and maximum for each variable. This reduced the training set to approximately 20,000 samples, which kept model file sizes small and training fast while retaining the information needed for daily min/max predictions.

Accuracy of Predictions

To evaluate the accuracy of the predictions, the coefficient of determination (R^2) was calculated for each of the four models against the training data. R^2 measures how well the model explains the variation in the target variable, where 1.0 is a perfect fit and 0.0 means the model performs no better than predicting the mean.

Max Temperature	Min Temperature	Max Humidity	Min Humidity
0.84	**0.84**	**0.68**	**0.82**
R^2 score	R^2 score	R^2 score	R^2 score

To provide a point-in-time comparison, the Hobart location was queried for January 1st across two separate years. Actual readings were extracted from the training dataset and compared against the model's predicted minimum and maximum values for those dates.

Location: Hobart	Actual Data				Predicted Data				Accuracy in 100%
	Min Temp	Max Temp	Min Humidity	Max Humidity	Min Temp	Max Temp	Min Humidity	Max Humidity	
Year 1 (Jan 1)	7.9°C	22.6°C	32%	95%	19.7°C	20.6°C	55.3%	60.5%	**45.5%**
Year 2 (Jan 1)	14.9°C	23.8°C	56%	94%	19.7°C	20.6°C	55.3%	60.5%	**79.4%**

The overall accuracy of the prediction is 62.5%.

The lower result for Year 1 is due to an unusually cold minimum temperature reading on that specific day, which was well below the seasonal average for early January. The model predicts the typical range for the season and location, so single-day outliers can affect point-in-time comparisons. The R^2 scores above reflect the model's performance across the full dataset.

Scope of Improvement:

Enhancing the accuracy of weather predictions is possible through several key strategies. One approach involves feature engineering — exploring additional relevant factors like wind speed, atmospheric pressure, and cloud cover to gain a more complete understanding of weather patterns. Advanced machine learning algorithms, such as decision trees, random forests, and neural networks, offer another avenue for improvement by capturing more complex relationships in the data.

Additionally, increasing dataset size, continuous monitoring, and feedback mechanisms can collectively improve the accuracy and reliability of weather prediction models. By adopting these strategies, future weather prediction models can deliver more precise predictions of temperature and humidity.